

STUDENT ADMISSION PREDICTION USING XGBOOST

End-to-End Machine Learning and Streamlit Deployment Project

1. Abstract

This project develops an AI-powered student admission prediction application using XGBoost. The system learns relationships between academic/profile attributes and the historical Chance of Admit target, evaluates the model on held-out data, and provides real-time predictions through Streamlit.

2. Problem Statement

Estimate graduate admission probability from GRE Score, TOEFL Score, University Rating, SOP, LOR, CGPA and Research Experience.

3. Dataset Description

Feature	Type	Description
GRE Score	Integer	Graduate Record Examination score
TOEFL Score	Integer	TOEFL score
University Rating	Integer	University rating from 1 to 5
SOP	Float	Statement of Purpose rating
LOR	Float	Letter of Recommendation rating
CGPA	Float	Undergraduate CGPA
Research	Integer	Research experience indicator
Chance of Admit	Float	Target value from 0 to 1

Dataset size after preprocessing: 500 rows. Missing values before cleaning: 0; remaining: 0.

4. Preprocessing

- Removed Unnamed: 0 and Serial No. because they are identifiers.
- Trimmed column-name whitespace.
- Converted modeling columns to numeric.
- Removed rows with missing required modeling values.
- Used an 80:20 train-test split with random_state=42.

5. Exploratory Data Analysis

EDA was used to understand the target distribution, the relationship between CGPA and admission chance, model feature importance, and prediction quality.

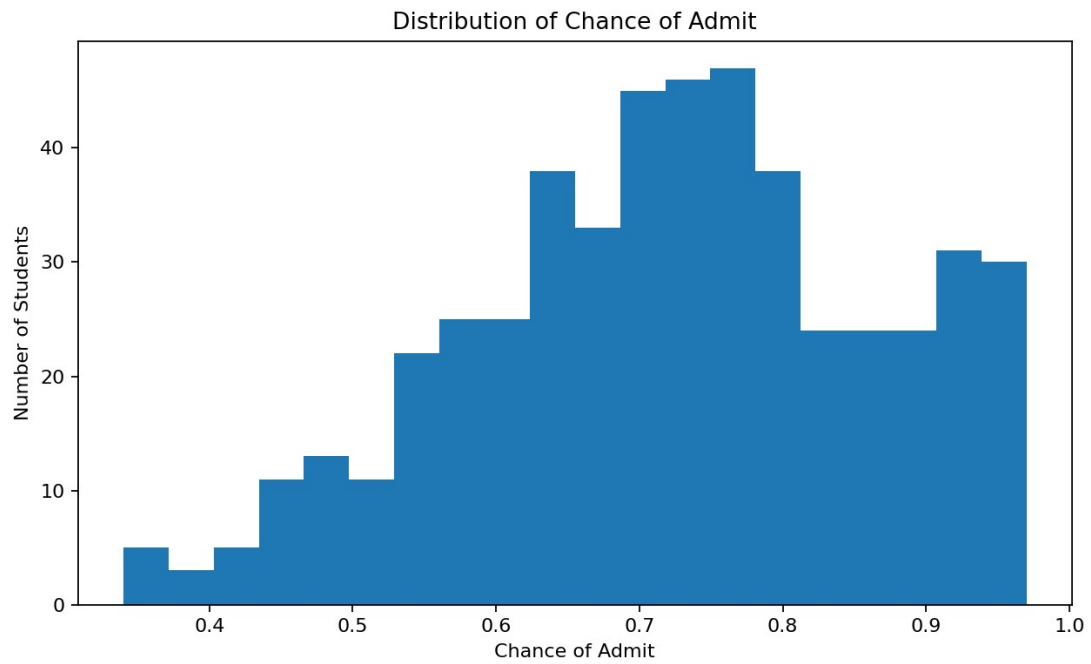


Figure 1. Target distribution

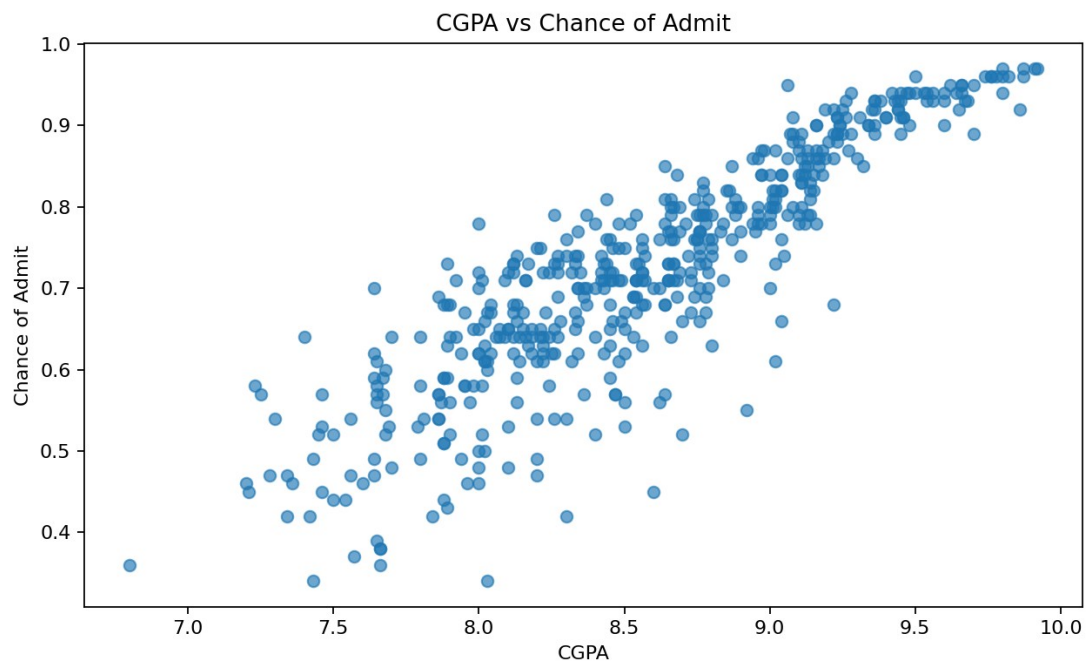


Figure 2. CGPA vs admission chance

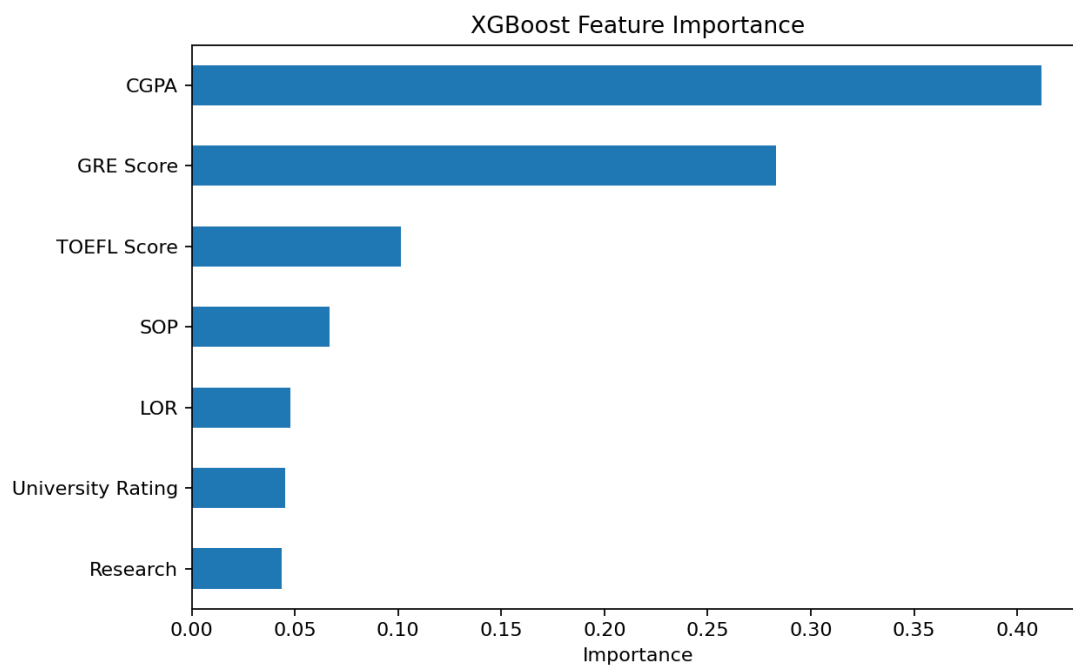


Figure 3. XGBoost feature importance

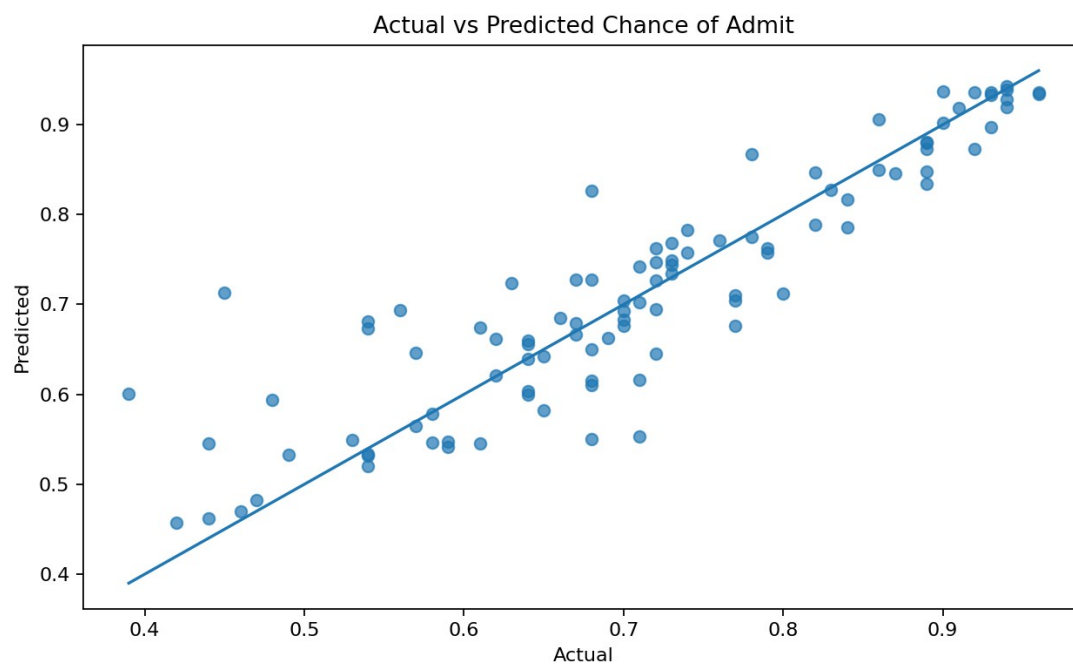


Figure 4. Actual vs predicted values

6. XGBoost Model

XGBoost Regressor was selected because it performs well on structured tabular data and can learn nonlinear relationships. Hyperparameters were selected using five-fold cross-validation.

Best parameters: {'colsample_bytree': 0.8, 'learning_rate': 0.02, 'max_depth': 3, 'n_estimators': 200, 'subsample': 0.8}

7. Model Evaluation

Metric	Test Result
MAE	0.0427
RMSE	0.0628
R ²	0.8071

MAE measures average absolute error; RMSE gives greater weight to larger errors; R² measures the proportion of target variance explained on the test set.

8. Streamlit Application

The interface accepts the seven model inputs, creates a one-row DataFrame in the same feature order as training, runs the saved model, clips the estimate to 0–1, and displays the result as a percentage.

9. Limitations

- The dataset is relatively small.
- Historical observations may not represent current admission policies.
- The prediction is an estimate rather than an official admission decision.
- Performance can change on unseen populations.

10. Conclusion

The project implements the complete machine learning lifecycle: preprocessing, EDA, XGBoost training and tuning, evaluation, model serialization, and Streamlit deployment.

11. Dataset Reference

Supplied dataset: data/graduate_admission.csv. Public reference:
<https://www.kaggle.com/datasets/mohansacharya/graduate-admissions>